

Certifying What Helps Customer-Return Timing: A Screen-and-Confirm Test for Conditioning Signals, and Why Decay Is Nearly Enough

Sang Su Lee
Thumbtack, Inc.
San Francisco, CA, USA
psulee@thumbtack.com

Shishir Dash
Thumbtack, Inc.
San Francisco, CA, USA
shishirdash@thumbtack.com

Vineeth Loganathan
Thumbtack, Inc.
San Francisco, CA, USA
vloganathan@thumbtack.com

Vijay Raghavan
Thumbtack, Inc.
San Francisco, CA, USA
vraghavan@thumbtack.com

Abstract

Practitioners enrich customer-return models with ever more signals (lifetime value, category, recency/frequency, calendar, geography), and the temporal-point-process (TPP) literature follows suit with covariate- and external-covariate-conditioned intensities. But *does any of it improve the timing*, and how would you know? A null (“feature X doesn’t help”) is only meaningful if the model *could* have found a signal. We make two contributions—a method and a measurement—to answer this credibly. (i) A **screen-and-confirm protocol** that *certifies* whether a candidate signal improves a TPP’s event-timing likelihood: a positive control plants a coupling of known strength and confirms the model recovers it, so a real-data null can be read as “no signal” rather than “weak method.” The control is validated for categorical and continuous encodings, and on a real clock-driven dataset (NYC taxi hour-of-day). (ii) A **model-free ceiling** quantifying how little of customer-return timing is *point*-predictable at all ($\leq 5\%$ of gap variance from any covariate; returns are near-memoryless). With these we certify a clean result on three public benchmarks (Amazon, Taobao, RetailRocket) and a real marketplace (Thumbtack): the inter-event clock—continuous-time *decay*, long known to beat frozen-intensity models—is nearly *sufficient*, and the conditioning the field keeps adding is redundant or harmful on top of it (statistically null on the public benchmarks, ≤ 0.06 NLL; null to mildly harmful on the marketplace). We do *not* claim to discover that decay helps; our contribution is the tools that turn “conditioning doesn’t help” into a checkable, certified statement—plus an honest-evaluation account of the read-out/leakage pitfalls we hit and retracted.

Keywords

temporal point processes, customer return, conditioning, positive controls, evaluation

1 Introduction

“When will this customer come back?” underlies notification timing, CRM prioritization, and lifecycle modeling. Temporal point processes (TPPs) answer it by modeling the conditional intensity

$\lambda^*(t \mid \mathcal{H}_t)$ —the instantaneous event rate given the past. Neural TPPs (RMTTP [1] $\rightarrow$ NHP [2] $\rightarrow$ Transformer-Hawkes/SAHP [3, 4] $\rightarrow$ state-space models [9]) have steadily improved likelihood fit. But a practitioner choosing a model for customer return faces a question the literature does not answer cleanly: **which components actually move return-timing, and how would you know?** Enriching event models with such signals is common practice: marked TPPs condition on event type/category by design [1, 2]; covariate TPPs add feature vectors and even learn their importance [13]; external-covariate TPPs inject seasonal/periodic drivers into the intensity [14]; and recency/frequency (RFM) and value features power deep churn and lifetime-value models [17]. These works typically *add* a signal and report a gain, in domains where it plausibly drives timing. We ask a different question for customer-*return* timing: not whether such a covariate *can* be added, but whether it adds anything once a strong inter-event backbone (decay) is already in place—and how to tell a true null from a method too weak to find the signal.

We answer empirically on three public benchmarks (Amazon, Taobao, RetailRocket) plus a real marketplace (Thumbtack), reporting temporal negative log-likelihood (NLL), inter-event RMSE, and MAE. Our contributions:

- **C1. A screen-and-confirm certification (method).** A protocol that certifies whether a candidate signal improves a TPP’s event-timing likelihood: a positive control plants a coupling of known strength and confirms recovery (categorical *and* continuous encodings, plus a real clock-driven dataset, NYC taxi), so a real-data null reads as “no signal,” not “weak method” (§5.5).
- **C2. A model-free ceiling (measurement).** How little of customer-return timing is *point*-predictable at all: $\leq 5\%$ of gap variance from any covariate; returns are near-memoryless (§5.4).
- **C3. The certified result.** Using C1–C2, on three public benchmarks and a real marketplace: continuous-time *decay* (the inter-event clock) is nearly *sufficient*, and the conditioning practitioners keep adding—LTV, category, RFM, calendar, geography—is redundant or harmful on top of it (statistically null on the public benchmarks, ≤ 0.06 NLL;

null-to-harmful, up to +0.65, on the marketplace), for a *dual mechanism* reason (§5.1, §5.3, §6).

The two contributions are one story. Our contribution is the *tools*—a screen-and-confirm certification (C1) and a model-free ceiling (C2); decay is a known mechanism, not our finding. The result (C3, decay nearly suffices) and the tools are inseparable: “conditioning doesn’t help” is trustworthy *only* because the positive control shows the null means “no signal,” not “a method too weak to find it.” The tool is what licenses the result. (Plain THP, our frozen-intensity reference, is an *ablation*, not a deployment target.)

The transferable lesson: for return-*timing*, the inter-event clock (decay) is the reliable lever, and **before believing that a feature helps, screen it against a positive control**—so that a null means “no signal in the data,” not “a method too weak to find it.”

2 Related Work

Neural TPP methods. RMTTP [1] → NHP [2] → attention models THP [3] and SAHP [4] → AttNHP [5]; intensity-free density models [6]; recent continuous-time state-space / latent-linear-Hawkes models (S2P2 [9]). EasyTPP [8] is the standard benchmark/codebase; the field evaluates on NLL, RMSE/MAE, and mark accuracy [10]. We compare these backbones plus conditioning components on the customer-return task.

Customer return / CLV. Buy-till-you-die models (BG/NBD [15]; Pareto/NBD [16]) and deep CLV [17] model value/return but not the full event-time intensity. Notably, BTYD models *assume* Poisson purchasing while alive—exponential, memoryless inter-purchase gaps per customer; our model-free ceiling (§5.4) independently confirms that forty-year-old assumption on four datasets, and the small residual previous-gap regularity (r^2 1–6%) is the mild timing regularity the Pareto/GGG extension models [22]. Hazard/point-process treatments of user return include Kapoor et al. [19] and Du et al. [20]; the closest neural treatment is the RNN survival model of Grob et al. [11], which predicts a *single* next return time per user. We instead model the full intensity λ^* and ask *which component matters on what data*, varying the backbone (NHP, THP, S2P2) and conditioning rather than committing to one architecture.

Conditioning TPPs on covariates. Injecting side information into the intensity is well studied: covariate TPPs encode feature vectors and learn per-feature importance (TransFeat-TPP [13]); external-covariate TPPs decompose seasonal/periodic drivers across temporal granularities (METP [14]); and marked TPPs [1, 3] condition on event type by construction. These report accuracy gains in settings where the covariate drives event dynamics. We do not propose a new conditioning mechanism; we measure whether the standard ones help for customer-return *timing* against a decay backbone, and add a positive-control screen so a null is interpretable.

Exogenous and causal TPPs. Estimating whether an *external* signal (calendar, geography, or an intervention such as marketing) drives event timing is, in general, a causal question; counterfactual/causal TPPs [12] formalize it but require interventional or carefully-controlled data. Our screen-and-confirm test (§5.5) is an *observational* necessary condition—a feature that fails to improve conditional likelihood when correctly supplied cannot be a useful timing signal—and we position the causal treatment as future work.

3 Models

We hold the conditioning framework fixed and vary the **backbone** and **components**.

Backbones. (i) *NHP*: continuous-time LSTM (CTLSTM); the hidden state decays between events, but the LSTM attenuates long-range history. (ii) *THP*: Transformer with causal self-attention over events; captures long-range dependence. In our *plain THP* ablation the intensity is fully *frozen* between events: we omit the original THP’s current-influence term $\alpha(t - t_j)/t_j$ [3], so published THP is only *state-frozen* (its intensity is linearly time-modulated between events) while our ablation removes between-event time-dependence entirely. (iii) *S2P2*: a state-space / latent-linear-Hawkes layer whose intensity integral is computed in *closed form* (no Monte-Carlo error).

Components (on the THP backbone). (a) *Decay head* (“-D”): let the hidden state decay between events, $h(t) = h(t_i) \exp(-\delta \Delta t)$ with a small learned δ ; this relaxes THP’s frozen-intensity assumption. (b) *LTV gate/shift* and (c) *category-aware decay*: condition the intensity on a per-customer value signal and event category. (d) *RFM*: a causal recency/frequency/cadence feature—frequency = $\log(1 + \text{position})$ and cadence = $\log 1p$ of the causal prefix-mean of inter-event times (using only events up to position i , to avoid leaking the target). (e) *Exogenous*: per-event season and region embeddings. (f) *Continuous covariate* (“-W”): a learned linear projection that injects a per-event continuous covariate vector into the event representation (we call it the *weather projection* after its original use case); it carries the continuous positive control and the taxi hour-of-day covariate (§5.5). We make *no* architectural-novelty claim; these are standard components whose *data-dependent value* we measure.

4 Experimental Setup

Datasets. Three public benchmarks—Amazon (product reviews, essentially one event type), Taobao (e-commerce actions, 17 types), RetailRocket (very long web-browsing sequences)—and **Thumbtack**, a U.S. home-services marketplace where a customer sends a *paid request* to professionals for a job (we keep customers with ≥ 3 such requests; a customer’s per-event “region” is their Designated Market Area, DMA—one of ~ 210 U.S. media-market regions). Public benchmarks use the EasyTPP-Gatech splits; Thumbtack is proprietary and reported in *relative* terms.

Metrics. Our primary metric is the **temporal NLL** ($\downarrow$). For an event-time sequence $t_1 < \dots < t_n$ with conditional intensity $\lambda^*(t) = \lambda(t \mid \mathcal{H}_t)$, the temporal log-likelihood is

$$\log L = \sum_{i=1}^n \log \lambda^*(t_i) - \int_{t_0}^{t_n} \lambda^*(u) du, \quad (1)$$

and we report the *per-event* temporal NLL = $-\frac{1}{n} \log L$. It isolates return-*timing*, is computed by one consistent pipeline across all datasets and model families (the integral in closed form for S2P2-F, by Monte-Carlo quadrature otherwise), and excludes the mark term. (Cross-model NLL therefore mixes a closed-form compensator for S2P2-F with an $n_{mc} = 10$ Monte-Carlo estimate for the others; we treat the S2P2-F-vs-MC NLL gap cautiously and rest our claims on the *same-estimator* decay contrast (THP vs. THP-D, an identical Monte-Carlo compensator).) We also report **inter-event**

RMSE/MAE ($\downarrow$); these are comparable *within a dataset only*, as the time unit differs across datasets. The same training protocol (Adam, warmup, early stopping on validation NLL) is used for all models, with mean $\pm$ std over seeds {42, 123, 456}. Throughout, we read a conditioning delta as *null* when $|\Delta| < 2\times$ its seed std and as real otherwise; the taxi recovery of \$5.5 ($\approx 3\times$) clears this bar, and no decay-regime conditioning delta does.

Backbone validation. Before drawing component conclusions we confirm our backbones reproduce known behavior: the relative ranking of S2P2, NHP, THP, and RMTTP matches the S2P2 paper [9]. (Three further reimplementations did not reproduce their published likelihoods in our pipeline; we omit them rather than report numbers we cannot trust, with details in App. A.)

5 Results

We present the evidence bottom-up. The decay gain (§5.1) and backbone comparison (§5.2) establish that the inter-event clock dominates; the conditioning redundancy (§5.3) and the **model-free ceiling** (§5.4, contribution C2) measure how little signal remains; and the **screen-and-confirm protocol** (§5.5, contribution C1) is the method that makes the negatives credible. The two framed contributions thus arrive last by design—the earlier subsections build to them.

5.1 Decay is the dominant timing lever (a known mechanism, quantified here)

Adding the decay head to a plain Transformer-Hawkes (THP $\rightarrow$ THP-D) yields large temporal-NLL gains on *every* dataset (Table 1): Amazon $0.32 \rightarrow -2.55$ ($\Delta -2.9$), RetailRocket $0.52 \rightarrow -3.53$ ($\Delta -4.0$), Taobao $-2.09 \rightarrow -2.58$ ($\Delta -0.5$); all mean over 3 seeds, std ≤ 0.09 (Table 2); and on Thumbtack a Δ of -1.26 . This gain is *distributional*—it sharpens the timing *likelihood*, not point error: with the point-prediction leak removed (§7), THP-D’s RMSE/MAE match plain THP on every dataset (§5.4). An earlier large MAE reduction we reported here was that read-out artifact and is retracted; the point-prediction view is the model-free ceiling of §5.4. The gain’s *size* varies (small on Taobao, large elsewhere), but its *sign* is negative on every dataset—this is the most consistent result in the study. Mechanism: decay relaxes THP’s frozen-between-events intensity so “time since last event” drives the rate—the classical mechanism of Hawkes processes and the neural Hawkes process [2, 18]. *Is the gain just decay, or THP?* Continuous-time models with built-in decay (NHP, and the state-space S2P2) likewise sit far below plain THP; the lever is the decay mechanism, not the backbone it is attached to. (Plain THP is a fair *ablation*—identical training and tuning, the between-event decay the single controlled variable—not a model one would deploy; modern attention TPPs are not frozen, and our plain THP omits the original’s current-influence term (§3), so the contrast quantifies the full between-event time-dependence channel rather than a deficit of published THP.) The same mechanism underlies S2P2’s strong likelihood in §5.2—exponential decay is the dominant mode of its state-space relaxation dynamics—so “decay is the lever” spans architectures, not merely the THP head.

Table 1: Decay gain: Δ temporal NLL from adding the decay head (THP $\rightarrow$ THP-D); negative = better. Mean over 3 seeds; Thumbtack relative. Negative on every dataset—size varies (smallest on Taobao), sign does not.

	Amazon	Taobao	RR	Thumbtack
Δ temporal NLL	-2.9	-0.5	-4.0	-1.26

5.2 No single best backbone—data decides on likelihood

This comparison is *not* itself a contribution; it plays two supporting roles—confirming our reproductions track the literature (Backbone validation, above) and locating the decay mechanism. Table 2 reports temporal NLL (mean $\pm$ std over three seeds) for the models we could seed cleanly. The robust, *same-estimator* comparison is the **decay contrast** (THP $\rightarrow$ THP-D, both Monte-Carlo): decay lowers NLL by whole nats on every dataset (std ≤ 0.09), confirming §5.1 with error bars. The closed-form state-space S2P2-F is competitive-to-best on the public sets (-3.44 Taobao, -4.95 RR), but its compensator is *exact* while the THP models’ is Monte-Carlo, so we parenthesize it as a reference and do *not* rank it against the MC models (its Thumbtack cell is unavailable). The Monte-Carlo S2P2 and NHP variants are omitted; single runs do not change the picture. We therefore make *no* universal-backbone-winner claim: decay is the *first-order* lever (whole nats), residual cross-backbone differences are *second-order* and data-dependent, and “sufficient” means decay reaches near the attainable ceiling—not that the backbone is irrelevant. This is consistent with the large-scale finding of Bosser and Ben Taieb [21] that time-NLL differences across neural TPP architectures are small; our multi-nat contrasts arise from the deliberately fully-frozen ablation (§3), not from cross-architecture spread. We also make no claim against IntensityFree/FullyNN/AttNHP (dropped; competitive in their own papers). (Apparent point-timing RMSE differences are largely a *read-out* artifact, not a backbone property; §7.)

Pre-specified sanity filter. We apply one fixed exclusion rule, decided from data/model properties rather than the ranking we wished to see: **implausible-given-inputs** variants are dropped—e.g. browse-augmented models reporting NLL -8 to -16 *including on Amazon, which contains no browse data*, so the “gain” cannot come from the claimed signal.

5.3 Conditioning is redundant under decay

Why test this at all? Because adding these signals is exactly what practitioners and the recent TPP literature do (§2 covers covariate-, marked-, and external-covariate-conditioned models); the screen and ceiling let us *certify* whether it actually pays for return-timing. We add each such signal on top of the backbone (Table 3). A *weak* (no-decay) baseline does benefit from category (CatOnly $\Delta -0.73$). But **once the decay head is present, every conditioning signal is null or harmful**: category worsens it (CatOnly-D $+0.47$; ExogCat-D $+0.65$, the worst), exogenous calendar/geography is null (ExogTHP-D $+0.012$), RFM collapses to ≈ 0 once its target leak is removed (§7), and proxy-LTV does not improve likelihood

Table 2: Temporal NLL ($\downarrow$), mean $\pm$ std over 3 seeds. Public absolute; Thumbtack relative to THP-D (Δ). The same-estimator decay contrast (THP vs. THP-D, both Monte-Carlo) is the robust comparison. † S2P2-F uses an exact closed-form compensator (parenthesized; *not* ranked against the MC models; Thumbtack cell unavailable). Monte-Carlo S2P2 and NHP are omitted.

Model	Amazon	Taobao	RR	tt1 $^\Delta$
THP	0.32 $\pm$ 0.00	-2.09 $\pm$ 0.00	0.52 $\pm$ 0.09	+1.26 $\pm$ 0.01
THP-D (THP+decay)	-2.55 $\pm$ 0.01	-2.58 $\pm$ 0.00	-3.53 $\pm$ 0.07	0.00
S2P2-Base-F (cf.) †	(-0.57 $\pm$ 0.00)	(-3.44 $\pm$ 0.00)	(-4.95 $\pm$ 0.00)	—

Table 3: Conditioning Δ temporal NLL relative to the same backbone ($\downarrow$ better; + is worse), Thumbtack (relative); mean over 3 seeds, $\pm$ the seed std of the Δ . The harmful decay-regime deltas carry large seed variance, consistent with the instability/overfitting reading (§5.3); ExogTHP-D is within seed noise. Conditioning helps only the weak baseline; under decay it is null or harmful.

Variant	Signal added	Δ temporal NLL
<i>over THP (no decay)</i>		
CatOnly	category	-0.73 $\pm$ 0.01
ExogTHP	season + region	+0.007 $\pm$ 0.005
<i>over THP-D (with decay)</i>		
CatOnly-D	category	+0.47 $\pm$ 0.11
RFM-D	recency/freq/cadence	-0.01 $\pm$ 0.01
LTV-CTPP-D	proxy LTV	+0.47 $\pm$ 0.10
ExogTHP-D	season + region	+0.012 $\pm$ 0.016
ExogCat-D	season+region+cat	+0.65 $\pm$ 0.17

(LTV-CTPP-D +0.47). **The same holds on the public benchmarks** (Table 4): under decay every signal moves temporal NLL by $\lesssim 0.06$, while the weak (no-decay) baseline still gains from category. We distinguish two regimes the table conflates: on public data the decay-regime effect is statistically *null* (redundancy—the signal carries no information the clock lacks); on Thumbtack adding category under decay is mildly *harmful* (+0.47), which we read as extra capacity overfitting in the absence of signal rather than as negative information (the harmful deltas’ large seed variance, ± 0.11 – 0.17 in Table 3, supports the instability reading). We therefore treat LTV/category/RFM/exogenous here as *measured components*, *not contributions*: under the signals available offline they are redundant with decay. (Whether *real* customer-revenue LTV helps remains open—the offline proxy ceiling is low by construction.) To be explicit about scope: this is a claim about the per-customer *timing likelihood conditional on the event history*—not about targeting, churn/propensity, mark/volume, or value models, where these same features remain informative (§5.5).

5.4 A model-free ceiling: returns are near-memoryless

Why does conditioning fail? Because, model-free, there is almost nothing to condition *on*. By a *model-free ceiling* we mean an upper bound on predictability computed *without* fitting any TPP—the fraction of gap variance a covariate explains in a simple regression—so

Table 4: Conditioning Δ temporal NLL on the *public* benchmarks, confirming the Thumbtack pattern: under decay every signal is $\lesssim 0.06$; the weak (no-decay) baseline still gains from category. Mean $\pm$ seed-std of Δ over 3 seeds; every decay-regime delta is within $\sim 1\sigma$ of zero (null by the §4 rule).

Δ NLL vs. same backbone	Amazon	Taobao	RR
CatOnly-D (+decay)	+0.03 $\pm$ 0.04	+0.00 $\pm$ 0.00	-0.06 $\pm$ 0.05
RFM-D (+decay)	+0.00 $\pm$ 0.08	-0.01 $\pm$ 0.00	-0.01 $\pm$ 0.03
CatOnly (no decay)	-0.32 $\pm$ 0.00	-0.02 $\pm$ 0.00	-1.70 $\pm$ 0.04

it bounds the *mean-shift (point-prediction)* signal available to any model. By the reconciliation argument below it does *not* bound distributional (likelihood) gains; the NLL-space redundancy claim rests on the measured deltas (Tables 3, 4) and the screen (§5.5), with the ceiling explaining why *point-prediction* gains, specifically, are unavailable. Table 5 shows that no available covariate explains more than a single-digit percentage of gap variance on *any* of the four datasets (we report the fraction of *log-gap* variance explained: r^2 for the continuous previous gap, one-way η^2 for categoricals): on Thumbtack the previous gap is the best single predictor at $\approx 2.2\%$ ($r = 0.147$), category $\approx 1.7\%$, and season and DMA region explain effectively nothing ($\eta^2 \approx 5\text{--}8 \times 10^{-4}$); on the public benchmarks the pattern repeats (previous gap 1.4–6.3%, event-type/mark $\eta^2 \leq 1.1\%$). Two regularities: the ceiling is single-digit everywhere, and the best single covariate is *always the previous gap*—i.e. the inter-event clock itself, sequence-derivable and already captured by decay. Nor is the ceiling low by construction: on a genuinely clock-driven dataset (NYC taxi, §5.5), the same model-free check on the exogenous hour-of-day covariate yields $\text{corr} = 0.15$ ($r^2 \approx 2\%$)—an order of magnitude above Thumbtack’s season η^2 —and the conditioning screen fires there; on customer-return data the seasonal signal simply is not present. Moreover, behavioral surge (a burst of sessions or a marketing touch) *appears to carry near-zero lead-time*—it coincides with the return rather than preceding it by days—so it could not be used to predict a return ahead of time (an informal observation, not a measured result; §7). Return timing is dominated by an exogenous, near-memoryless latent need; the inter-event clock captured by decay is most of the recoverable signal. Precisely, the process is near-*renewal*: the hazard depends strongly on time since the last event (that is what decay captures) but on little else—“near-memoryless” throughout means memoryless *beyond the current gap’s clock*, not a constant hazard.

Table 5: Model-free fraction of log-gap variance explained by each covariate (r^2 for the continuous previous gap; one-way η^2 for categoricals). Single-digit everywhere, and the best covariate is always the *previous gap*—the clock that decay captures. Public columns: pooled splits, category = event type (marks); season/region covariates are unavailable on the public benchmarks.

Covariate	Amazon	Taobao	RR	Thumbtack
Previous gap (r^2)	1.4%	3.4%	6.3%	$\approx 2.2\%$
Category/mark (η^2)	1.1%	0.4%	0.3%	$\approx 1.7\%$
Season (month, η^2)	—	—	—	$\approx 5 \times 10^{-4}$
Region (DMA, η^2)	—	—	—	$\approx 8 \times 10^{-4}$

Reconciling with the decay gain (§5.1). A natural objection: if returns are near-memoryless, how can decay lower NLL by 2–3 nats? Because the two measure different things. The ceiling is *point-prediction* variance—how much a covariate moves the conditional *mean* gap. The decay gain is a *distributional* (likelihood) improvement: even when the mean is nearly unpredictable, modeling the *shape* of the intensity (the post-event refractory dip and its drift back, and the compensator $\int \lambda^*$) calibrates the timing density far better than a flat-rate baseline. Decay does not predict *which* gap better (low r^2); it places probability mass over *when* far better (large Δ NLL). So the findings are consistent: little is point-predictable, yet correct temporal *calibration* is worth a lot—and it comes from the inter-event clock, not from external covariates. To make this concrete: on a *perfectly* memoryless process (i.i.d. $\text{Exp}(\lambda)$ gaps, where *no* covariate has any predictive power), a model whose rate is off by a constant factor c pays $c - 1 - \ln c$ excess nats per event—1.6 at $c = 4$, 6.7 at $c = 10$. Multi-nat NLL gains are thus attainable at *zero* point-predictability: getting the rate *scale/shape* right (what decay does, versus THP’s frozen between-event rate) is worth nats even when *which* gap is unpredictable. Empirically this is exactly what we observe: once the point-prediction leak is removed (§7), THP, THP-D, and LTV-CTHP-D sit at the *same* inter-event RMSE (Amazon ≈ 0.30 , Taobao ≈ 0.14 , RetailRocket ≈ 9.5), indistinguishable from predicting the global-mean gap, and a plain RFM→gradient-boosting regressor matches the neural TPPs on this point-prediction. Nothing beats the constant on *which* gap, while decay still wins whole nats on *when*—the point-prediction ceiling and the distributional decay gain are two faces of the same near-memoryless process.

5.5 Screening exogenous features: synthetic recover, real reject

The protocol. (1) Fix the candidate feature’s *encoding* and the prediction target (here, inter-event timing). (2) On synthetic data, plant a feature→target coupling of tunable strength β and confirm the conditioned model recovers it *monotonically* in β —a positive control that calibrates sensitivity. (3) Run the *identical* pipeline on the real feature. (4) Interpret *within the control’s scope*: given a passing control, a flat real result certifies “no signal in that encoding,”

not a weak method. The pattern ports standard positive/negative-control practice—placebo and refutation tests in econometrics, sanity checks in ML [23]—to TPP conditioning.

A null is only informative if the method *could* have found a signal. We therefore validate the exogenous pipeline with a **positive control**: we generate synthetic sequences in which an exogenous field (season) is coupled to the next gap at a tunable strength β (a causal coupling that a decay-only model cannot recover, since the field is i.i.d. across events), and check that a season/region-conditioned model recovers it. It does, *monotonically* in β (Fig. 1): ExogTHP improves by $-0.059, -0.232, -0.824$ at $\beta = 0.5, 1, 2$ (and $+0.001$ at $\beta = 0$, correctly null); the decay variant ExogTHP-D recovers as well ($-0.042, -0.143, -0.337$). The protocol is *not* specific to categorical features: replacing the categorical season/region embedding with a learned linear projection of a per-event *continuous* covariate z , and planting a continuous coupling gap $\sim \text{Exp}(e^{\beta z})$ with $z \sim \mathcal{N}(0, 1)$, yields the same monotonic recovery under decay— Δ time-NLL $+0.000, -0.092, -0.258, -0.738$ at $\beta = 0, 0.5, 1, 2$. Both instantiations are correctly null at $\beta = 0$. The same pipeline, applied to real calendar/geography (Thumbtack), yields no improvement (Table 3).

The real null is genuine, not a plumbing artifact. We verified that the exogenous signal reaches the model with rich variation: Thumbtack carries all 12 calendar months and 207 DMA regions (99.8% non-zero), well-distributed, with loader keys matching the data. Because the positive control confirms “signal present $\Rightarrow$ recovered,” the real-data null means “nothing to recover”—calendar and geography do not move the per-customer conditional gap. This screen-and-confirm pattern is the recommended way to read a conditioning null.

Scope of the claim. The screen adjudicates a *specific* pair: the candidate feature *as encoded* and the chosen target (here, inter-event timing). A null therefore means “no signal in that encoding for that target,” bounded by what the positive control shows recoverable—not “no exogenous signal of any kind.” Likewise, of the signals in C3, calendar/geography passed through this planted-coupling screen; the LTV/category/RFM nulls are *measured* under the same pipeline and bounded by the same ceiling but were not separately screened—per-pathway positive controls are future work. This boundary is consistent with the broader literature: on canonical weather-sensitive data the exogenous covariate drives *volume and marks* rather than per-event timing. As a model-free check, Beijing PM2.5 alert *onsets* are near-uniform across calendar months and their inter-onset gaps are essentially season-invariant, and on the Walmart weather dataset temperature is uncorrelated with the inter-sale gap ($r = 0.005$); a *timing* screen is thus correctly null there, even though a mark/volume model benefits from the same covariate [13, 14]. Extending the screen to continuous and lagged encodings (each with its own positive control) is a direct generalization.

A real positive control. The screen also fires on a *real* exogenous signal where one genuinely exists. On NYC green-taxi pickups (Jan 2019), inter-pickup gaps are modulated by *hour-of-day* (rate high at rush hour, low overnight; model-free $\text{corr}(\sin \text{hour}, \log \text{gap}) = 0.15$). Encoding hour as a continuous cyclic covariate, the screen lowers temporal NLL by $\Delta = -0.025$

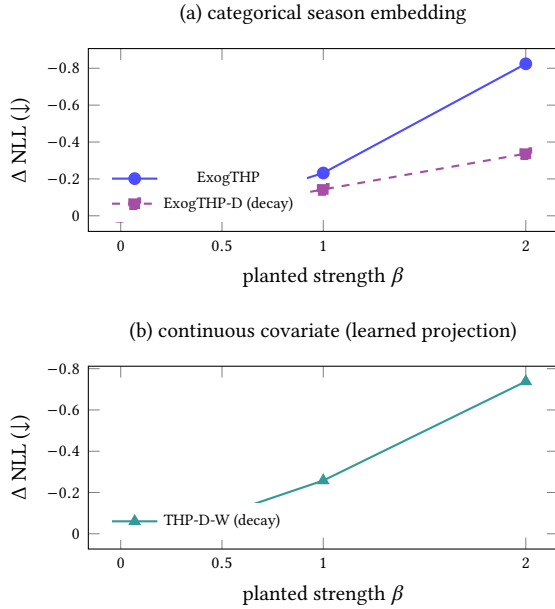


Figure 1: Positive control (two encodings). A planted exogenous→gap coupling of strength β is recovered *monotonically* in both: (a) a *categorical* season embedding (no-decay ExogTHP and decay ExogTHP-D), and (b) a *continuous* covariate via a learned linear projection (decay THP-D-W). $\beta = 0$ is correctly null in both, so the screen is not categorical-specific. This is what a *real* signal would look like—real calendar/geography (Table 3) does not.

(1.756 $\rightarrow$ 1.731; $n = 3$, well outside seed noise ± 0.008 ; replicated on a second month, Feb 2019, $\Delta = -0.021$; hour is genuinely exogenous—each sequence starts at its zone-day’s first pickup, so cumulative time does not encode the clock hour)—a *modest* recovery that tracks the *modest* real signal, in contrast to the strong synthetic β and the flat (≈ 0) real calendar/geography. This closes the loop: synthetic controls calibrate sensitivity, real negatives (PM2.5, Walmart) and a real positive (taxi) confirm the screen discriminates on real data—so the customer-return null is a property of that data, not a weakness of the method. We still describe screen-and-confirm as a *validation discipline* (establish recoverability, then trust the null) rather than a turnkey discovery method.

6 Mechanism: why decay absorbs the rest

The redundancy has a *dual mechanism*. (A) **Sequence-derivable signals are already captured.** RFM and category are deterministic functions of the observed event history, which the decay backbone already encodes; supplying them explicitly adds no information (RFM-D ≈ 0 , CatOnly-D worse once decay is in). (B) **Exogenous signals do not move the conditional gap.** Season and region shift *demand composition* (who buys what, where) but not the per-customer *timing* of the next return, which is driven by an exogenous latent need realized same-day (§5.4). Timing $\neq$ demand-composition.

Together these explain why a single inter-event-clock mechanism (decay) is sufficient and the usual enrichments are redundant.

What this buys the practitioner. For the notification-timing and CRM use cases of §1: a decay-calibrated intensity supports return-*window* estimation (when the hazard recovers from the post-event refractory dip) and send suppression during the dip; it does *not* support point-timed sends (point prediction sits at the global-mean ceiling, §5.4), and the effect of *timing an intervention* is a causal question requiring interventional data (§2, future work).

7 Limitations and honest evaluation

We report results we initially got wrong and corrected, in the spirit of honest evaluation. An NHP target-indexing bug once produced a fake $7\times$ MAE improvement; an RFM cadence feature once leaked the target through a full-sequence normalizer; an S2P2 RMSE was read from an untrained auxiliary head; and the decay head’s point-prediction read the hidden state *after* decaying it by the actual next inter-event interval, leaking the very gap it predicts (a counterfactual that alters only the held-out gap moves the prediction in lock-step, $\text{corr} \approx 1$)—which had inflated the decay models’ RMSE/MAE and the apparent point-timing decay gains. Each was caught by sanity-checking predictions against targets; we fixed the decay read-out to use the pre-decay (history-only) state, added a counterfactual regression test, and report only the corrected, leak-free numbers, which show *decay-flat* RMSE/MAE. Temporal NLL—a density evaluated at the observed event times and treated identically across all models—is unaffected by these point-prediction read-out bugs. Separately, in earlier ranking experiments a TPP’s apparent ranking ability was dominated by the score *read-out* (intensity integral vs. a learned time head) and the held-out *anchor* rather than the architecture; an apparent “likelihood-vs-ranking decoupling” did not survive re-validation, so we make *no* ranking claim. Scope limits: one real dataset, reported only in relative terms; the customer filter (≥ 3 paid requests) conditions on repeat return, so seasonality acting on acquisition or the *first* return is excluded by construction; proxy-LTV understates real-LTV conditioning (open); and our S2P2 point-timing is read from an auxiliary head, so we do not treat its RMSE as representative of the method. Three claims rest on argument more than on a dedicated experiment, which we flag: the model-free *season/region* ceilings (§5.4) are measured on Thumbtack only (the public benchmarks carry no calendar/geography covariates), and platforms with strong external seasonality may sit under a higher ceiling, so the one-day ceiling-plus-screen check should be re-run per dataset instead of assuming our $\lesssim 5\%$ figure; the “decay already encodes RFM/category” mechanism (§6) is *inferred*, not probed (no representation-probing experiment); and the “surge has zero lead-time” observation is from exploratory analysis without a dedicated figure.

8 Conclusion

Two tools, a model-free ceiling and a screen-and-confirm certification, let us verify a clean result for customer-return *timing*: continuous-time decay (a known mechanism) is nearly *sufficient*, and once it is present, the conditioning practitioners add is redundant or harmful. A model-free analysis shows why: returns are

near-memoryless, with $\lesssim 5\%$ of gap variance explainable (and, informally, near-zero surge lead-time). Because a conditioning null is only meaningful against a positive control, we recommend a **screen-and-confirm** protocol: confirm the pipeline recovers a planted synthetic signal, then trust the real-data null. Future work: a causal/counterfactual TPP treatment of *interventions* (e.g. marketing), which needs interventional or randomized-holdout data; representation-probing (or mutual-information) experiments on the decay backbone’s hidden state, to turn the *inferred* “decay already encodes RFM/category” mechanism of §6 into a measured one; and generalizing screen-and-confirm into automatic exogenous-feature discovery.

A Reproducibility details

Splits. Public benchmarks use the EasyTPP-Gatech [8] train/dev/test splits; Thumbtack uses a chronological per-customer split, reported only in relative terms. **Optimization (all models).** Adam, learning rate 10^{-3} , linear warmup over 5 epochs then Reduce-LROnPlateau (factor 0.5, patience 5, min lr 10^{-5}), gradient clipping at 1.0, up to 50 epochs with early stopping (patience 10) on validation temporal NLL. Hidden size $d_{\text{model}} = 64$; batch size 64 (RetailRocket 16, sequences truncated to 512). $\pm$ is mean/std over seeds {42, 123, 456}. **Backbones.** THP: causal self-attention, intensity constant between events (the original’s current-influence term $\alpha(t - t_j)/t_j$ is omitted; §3); “-D” adds the per-dimension hidden-state decay head (small learned δ). NHP: continuous-time LSTM. **S2P2 is our reimplementation** of a continuous-time state-space / latent-linear-Hawkes layer ($d_{\text{state}} = 64$, 2 layers); -F computes $\int \lambda^*$ in closed form ($n_{\text{mc}} = 0$), the non-F variant uses $n_{\text{mc}} = 10$. We do not use the original authors’ code. **Dropped baselines.** IntensityFree [6], FullyNN [7], and AttNHP [5] reimplementations did not reproduce published likelihoods in our pipeline (e.g. Taobao IntensityFree LL -1.43 vs. $+1.318$ reported), so we exclude their numbers and cite the originals, which report competitive results. **Synthetic positive control.** Per-event season $\sim \text{Unif}\{1..12\}$ i.i.d.; $\text{gap}_{j+1} \sim \text{Exp}(r_0 \exp(\beta \sin(2\pi \text{season}_j/12)))$ using the *source* event’s season (causal); region is an i.i.d. noise control. A pre-training validation confirms the planted correlation grows with β . The continuous variant draws $z \sim \mathcal{N}(0, 1)$ per event and sets $\text{gap}_{j+1} \sim \text{Exp}(e^{\beta z_j})$, fed through the weather projection. **Model-free ceiling (public columns of Table 5).** Computed over pooled train/dev/test splits, on log of positive inter-event gaps: r^2 from the Pearson correlation of adjacent within-sequence gap pairs ($\log g_j, \log g_{j+1}$); η^2 from a one-way decomposition of $\log g_{j+1}$ grouped by the source event’s type (ceiling_public.py). Applied to the Thumbtack event data, the same script yields $r^2 \approx 2.0\%$ and category $\eta^2 \approx 1.1\%$, consistent in magnitude with the cohort-derived Thumbtack column of Table 5. **Real positive control (taxi).** NYC green-taxi pickups, Jan 2019; sequences are per (pickup-zone, day) with ≥ 20 pickups (capped at 200, $n = 4000$); the per-event covariate is hour-of-day encoded as $[\sin(2\pi h/24), \cos(2\pi h/24), 0]$ through the same weather projection. Events are single-type (timing only).

References

- [1] N. Du et al. Recurrent Marked Temporal Point Processes. *KDD*, 2016.
- [2] H. Mei and J. Eisner. The Neural Hawkes Process. *NeurIPS*, 2017.
- [3] S. Zuo et al. Transformer Hawkes Process. *ICML*, 2020.

- [4] Q. Zhang et al. Self-Attentive Hawkes Process. *ICML*, 2020.
- [5] C. Yang, H. Mei, J. Eisner. Transformer Embeddings of Irregularly Spaced Events and Their Participants (AttNHP). *ICLR*, 2022.
- [6] O. Shchur et al. Intensity-Free Learning of Temporal Point Processes. *ICLR*, 2020.
- [7] T. Omi, N. Ueda, and K. Aihara. Fully Neural Network based Model for General Temporal Point Processes. *NeurIPS*, 2019.
- [8] S. Xue et al. EasyTPP: Towards Open Benchmarking Temporal Point Processes. *ICLR*, 2024.
- [9] Y. Chang, A. Boyd, C. Xiao, T. Kass-Hout, P. Bhatia, P. Smyth, and A. Warrington. Deep Continuous-Time State-Space Models for Marked Event Sequences. *NeurIPS*, 2025.
- [10] O. Shchur et al. Neural Temporal Point Processes: A Review. *IJCAI*, 2021.
- [11] G. Grob et al. A Recurrent Neural Network Survival Model: Predicting Web User Return Time. *ECML PKDD*, 2018.
- [12] K. Noorbakhsh and M. Gomez-Rodriguez. Counterfactual Temporal Point Processes. *NeurIPS*, 2022.
- [13] Z. Meng, B. Li, X. Fan, Z. Li, Y. Wang, F. Chen, and F. Zhou. TransFeat-TPP: An Interpretable Deep Covariate Temporal Point Processes. *ECAI*, 2024.
- [14] B. Li, L. Zhang, F. Tsung, and X. Zhang. METP: Multi-Granularity Integration of External Covariates for Temporal Point Processes. *AAAI*, 2026.
- [15] P. Fader, B. Hardie, K. Lee. “Counting Your Customers” the Easy Way: BG/NBD. *Marketing Science*, 2005.
- [16] D. Schmittlein, D. Morrison, R. Colombo. Counting Your Customers: Pareto/NBD. *Management Science*, 1987.
- [17] X. Wang, T. Liu, J. Miao. A Deep Probabilistic Model for Customer Lifetime Value (ZILN). arXiv:1912.07753, 2019.
- [18] A. G. Hawkes. Spectra of Some Self-Exciting and Mutually Exciting Point Processes. *Biometrika*, 1971.
- [19] K. Kapoor, M. Sun, J. Srivastava, and T. Ye. A Hazard Based Approach to User Return Time Prediction. *KDD*, 2014.
- [20] N. Du, Y. Wang, N. He, J. Sun, and L. Song. Time-Sensitive Recommendation from Recurrent User Activities. *NeurIPS*, 2015.
- [21] T. Bosser and S. Ben Taieb. On the Predictive Accuracy of Neural Temporal Point Process Models for Continuous-time Event Data. *TMLR*, 2023.
- [22] M. Platzer and T. Reutterer. Ticking Away the Moments: Timing Regularity Helps to Better Predict Customer Activity. *Marketing Science*, 2016.
- [23] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim. Sanity Checks for Saliency Maps. *NeurIPS*, 2018.